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CS 621 DATA ANALYTICS

Spatial Data

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SPATIAL DATA

- Data associated with specific geographical location.
- Location/Area names:
  - Indiana, US
  - Muncie, IN
  - Ball State University, Muncie, IN, US
- GPS coordinates: Latitude, Longitude and Elevation:
  - 40.206, -85,407, 936 feet

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SPATIAL DATA EXAMPLE 1

State	Total Votes
Alabama	2,323,282
Alaska	359,530
Arizona	3,387,326
Arkansas	1,219,069
California	17,500,881
Colorado	3,256,952
Connecticut	1,824,280
Delaware	504,010
District of Columbia	344,356
Florida	11,067,456
Georgia	4,997,716

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SPATIAL DATA EXAMPLE 2

longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
-122.23	37.88	41	880	129	322	126	8.3252	452600	NEAR BAY
-122.22	37.86	21	7099	1106	2401	1138	8.3014	358500	NEAR BAY
-122.24	37.85	52	1467	190	496	177	7.2574	352100	NEAR BAY
-122.25	37.85	52	1274	235	558	219	5.6431	341300	NEAR BAY
-122.25	37.85	52	1627	280	565	259	3.8462	342200	NEAR BAY

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CHOROPLETH MAP

- https://en.wikipedia.org/wiki/Choropleth_map
- A thematic map in which areas are shaded in proportion to some statistical measurement.
- A visualization across a geographic area, e.g., on a US map.

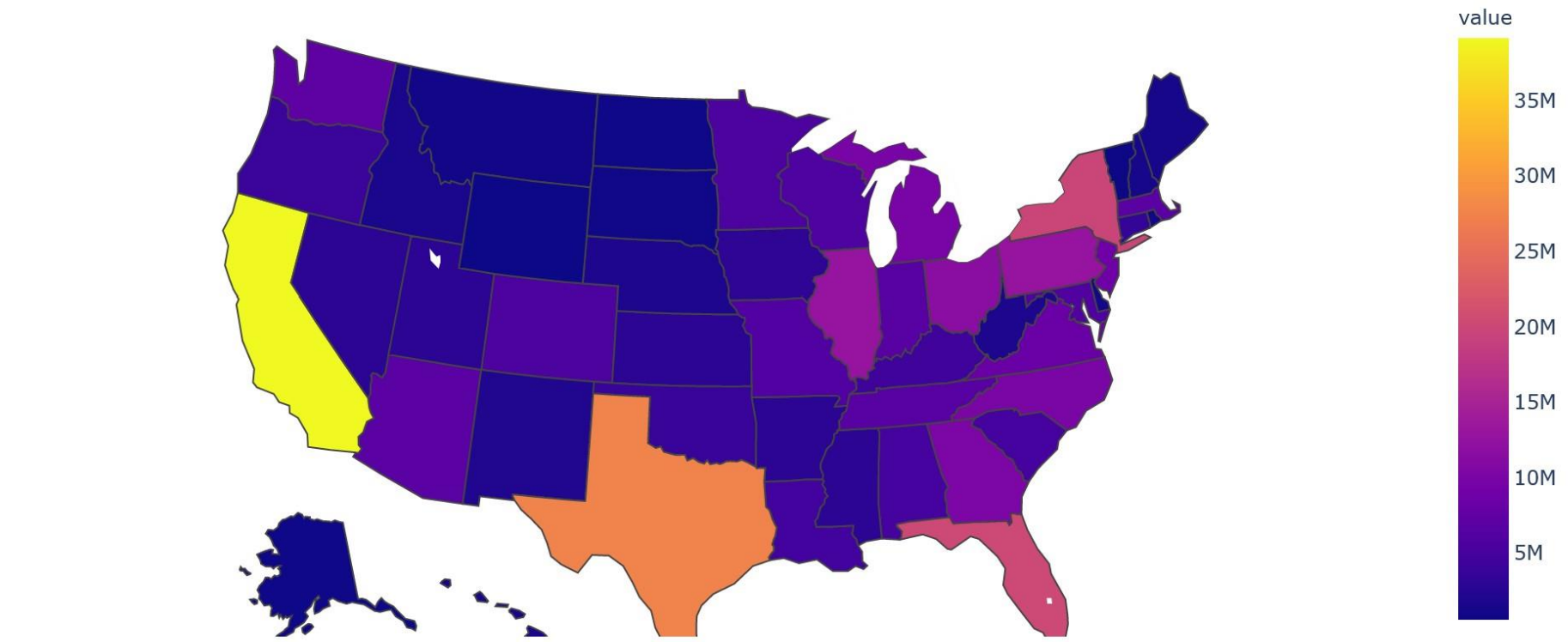
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CHOROPLETH MAP LIBRARIES

- Use Plotly: https://plotly.com/python/choropleth-maps/
- Use Folium - https://python-graph-gallery.com/292-choropleth-map-with-folium/

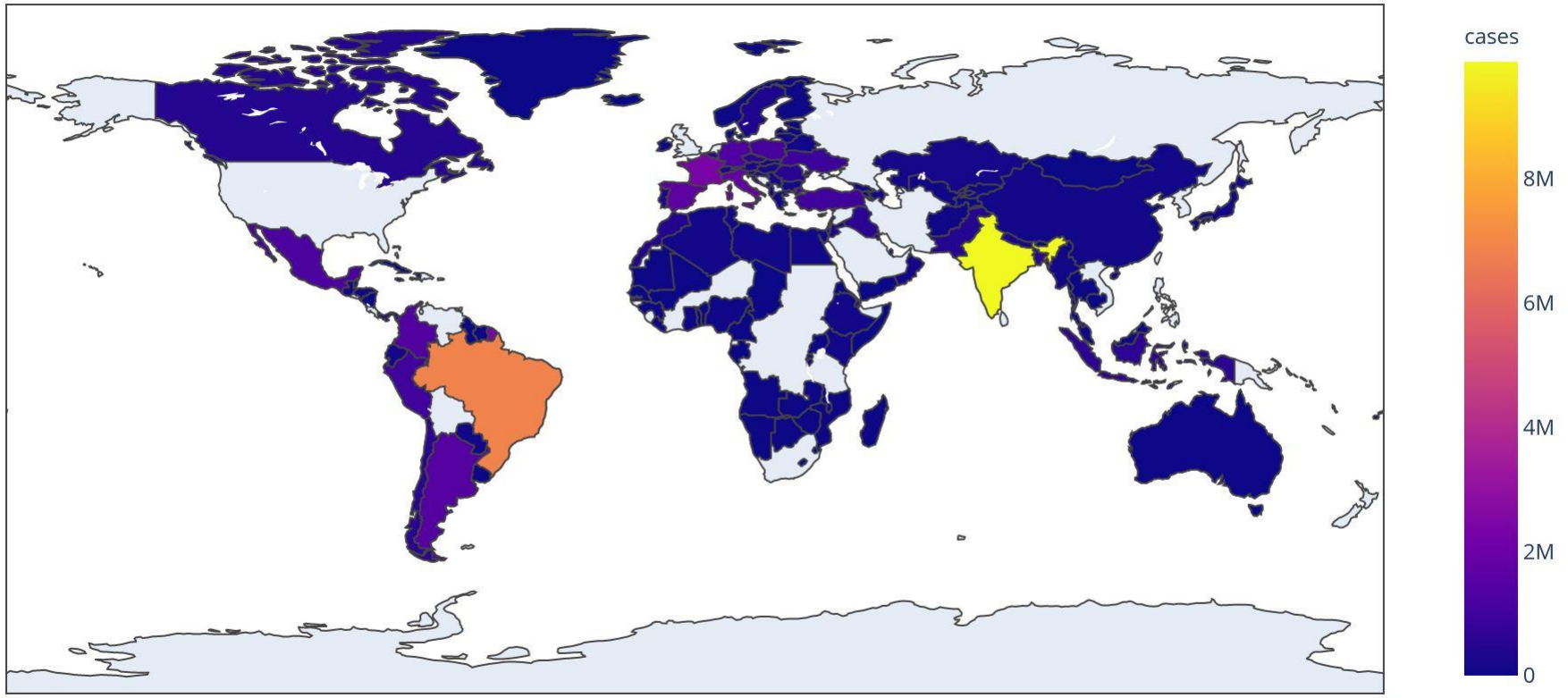
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STATE-LEVEL US POPULATION



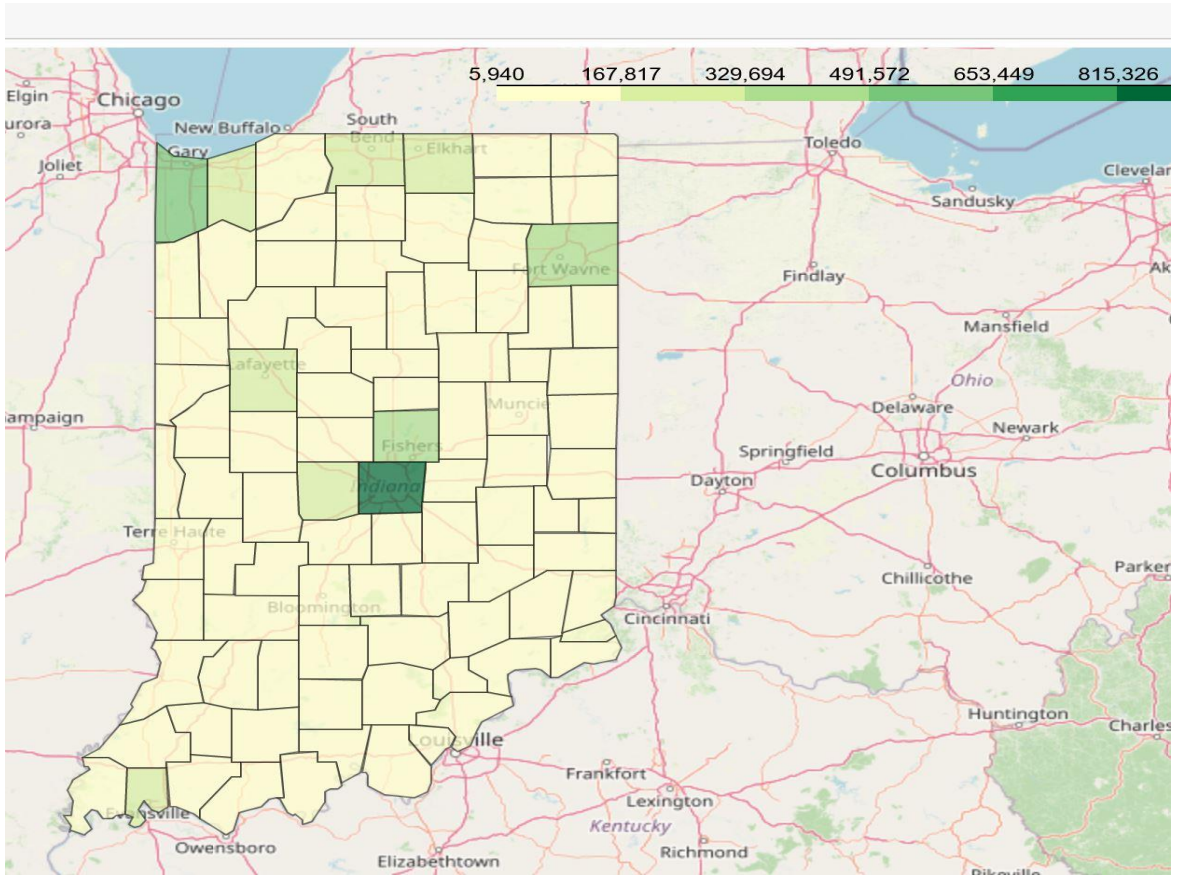
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WORLDWIDE CHOROPLETH MAP



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INDIANA STATE CHOROPLETH MAP



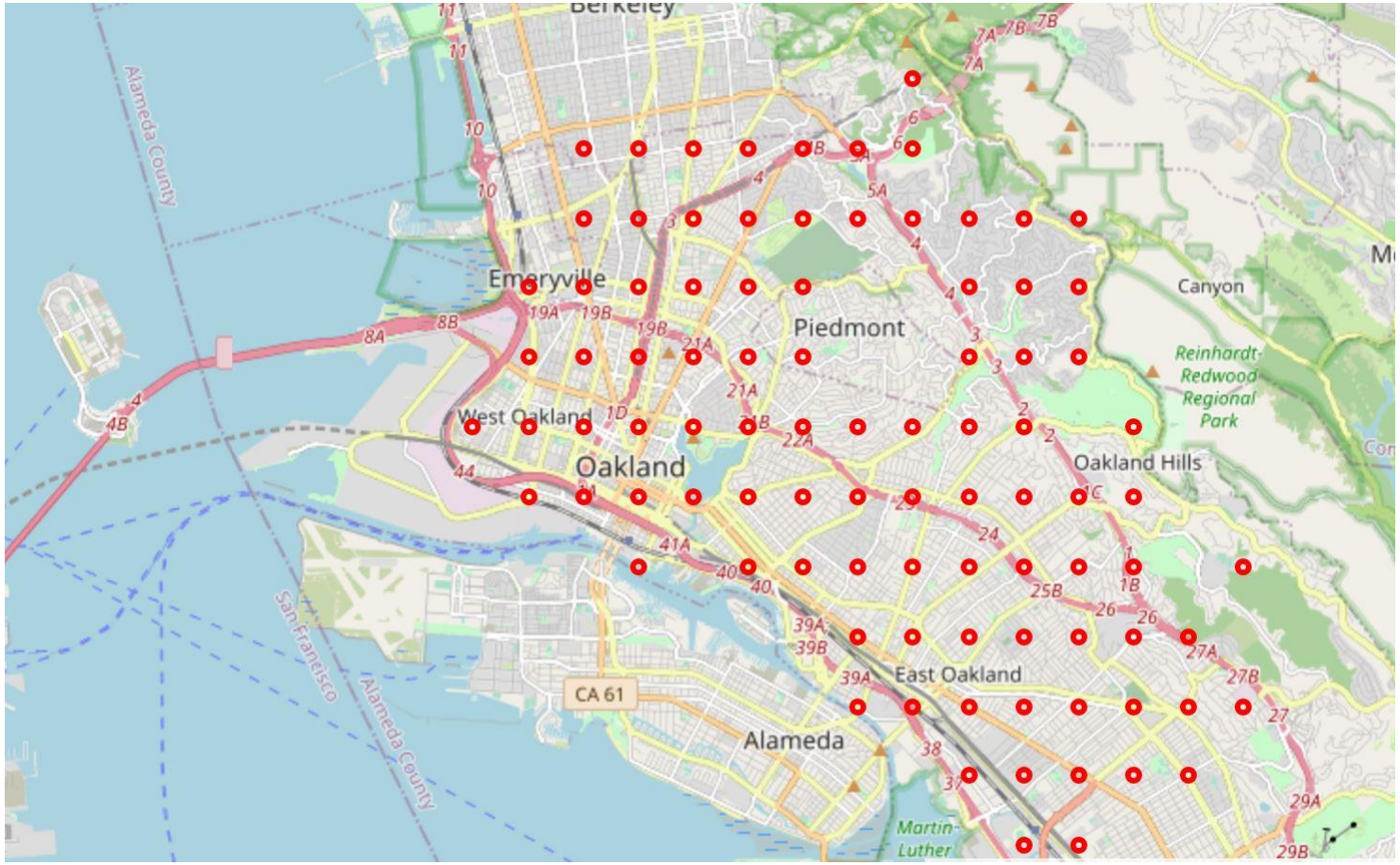
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CA. HOUSING GPS DATA

	longitude	latitude
0	-122.23	37.88
1	-122.22	37.86
2	-122.24	37.85
3	-122.25	37.85
6	-122.25	37.84
8	-122.26	37.84
10	-122.26	37.85
16	-122.27	37.85
19	-122.27	37.84
25	-122.28	37.85

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GPS DATA VISUALIZATION



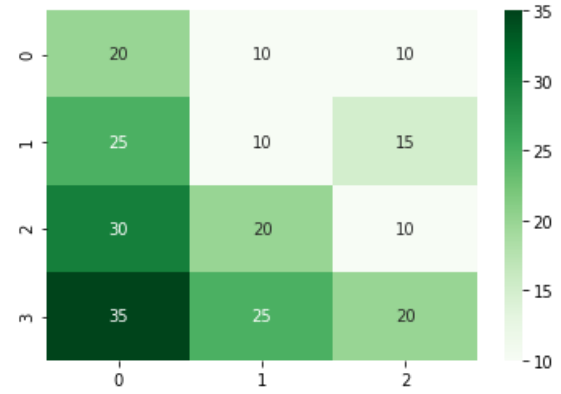
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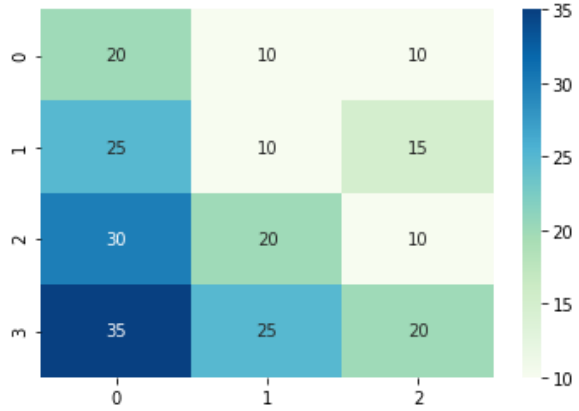
HEAT MAPS

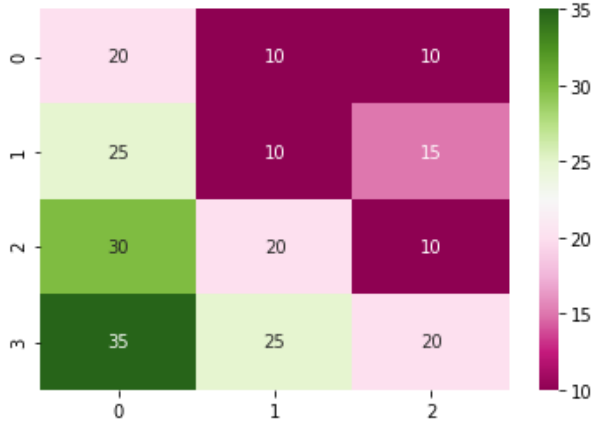
- Data by colors
- A data matrix
- Each value in the matrix will be presented by a color
- Intensity of color/shade: lighter ← … → darker
- Seaborn.heatmap has powerful support for drawing heatmaps. See class code demonstration for examples.

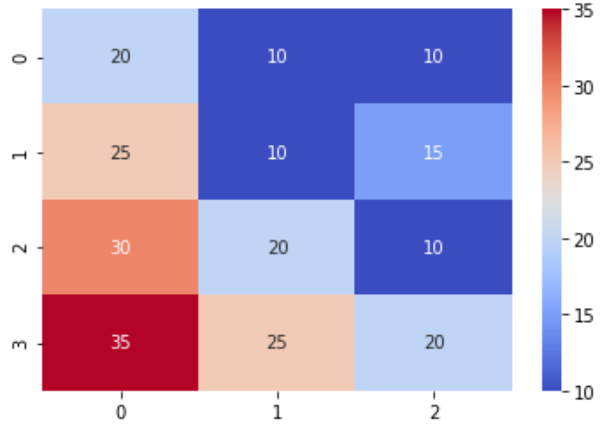
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HEATMAPS



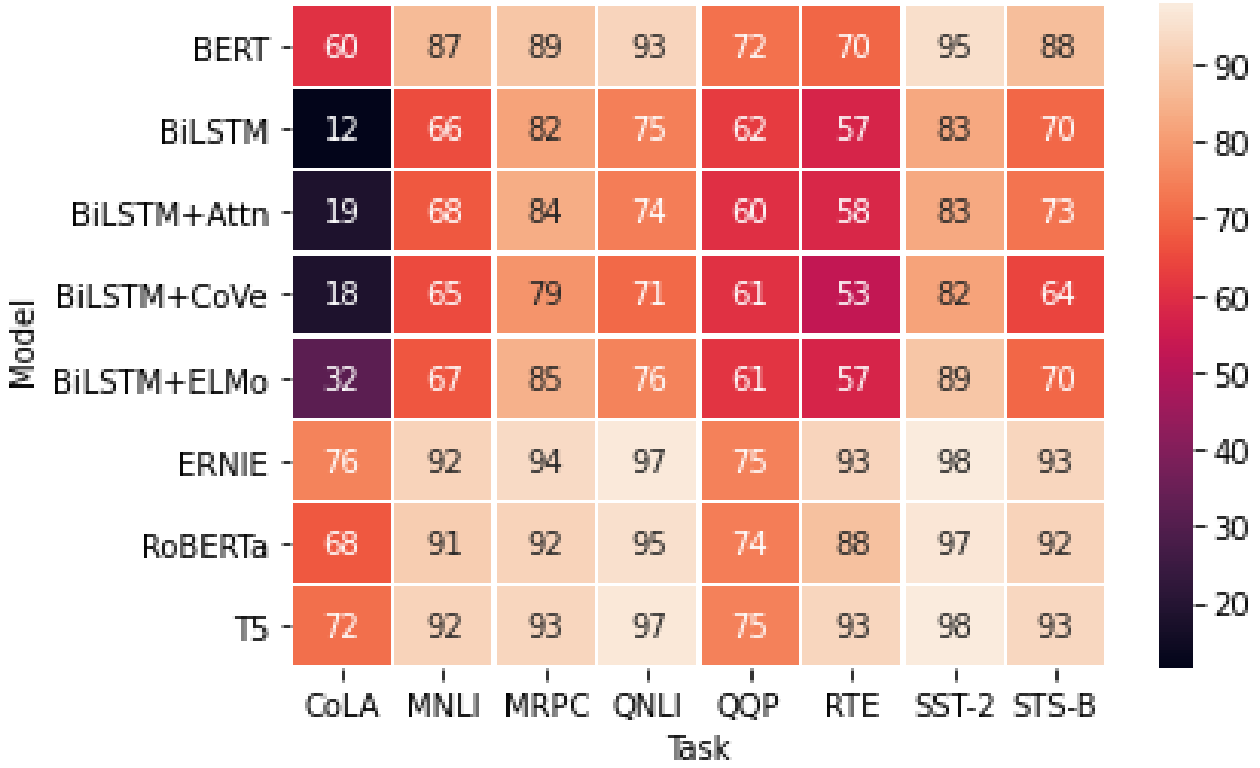






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HEATMAP





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WE FLY